Chapter 4 Quiz — Energy Transfer

Grade 4 · Energy Transfer and Transformation

Name: _____ Date: _____

Circle or write the best answer.

PART 1 — Multiple Choice

1. Energy transfer means energy ___.

a) disappears b) moves from place to place or object to object c) changes form d) stops

2. What is the best evidence that energy transferred?

a) a guess b) an observable change in a target c) the energy itself d) a prediction

3. A cup is tapped, and a paper target moves even though nothing touched it. This is evidence that ___ transferred energy.

a) sound b) heat c) light d) electricity

4. A flashlight lights up a card across the room. This is evidence that ___ transferred energy.

a) sound b) light c) heat d) motion

5. Warm water is placed next to a cool cup. The water cools and the cup warms. This is evidence that ___ transferred energy.

a) sound b) light c) heat d) electricity

6. A battery is connected to a bulb by wires, and the bulb lights up. This is evidence that ___ transferred energy.

a) electric current b) sound c) light d) heat

PART 2 — Diagram

7. Label the energy transfer path for a flashlight shining on a card.

Write SOURCE, PATH, and TARGET on the correct boxes.

[flashlight]	[-----> light ----->]	[card]

PART 3 — Data

8. A student places a cool metal cup in warm water and records the temperature.

Material	Start (°C)	Later (°C)
Warm water	38	34
Cool metal cup	21	27

Which material got warmer? Which got cooler? What does this show about energy?

PART 4 — Transfer / Transformation / BOTH

TRANSFER = energy moves. TRANSFORMATION = energy changes form. BOTH = moves AND changes form.

9. A flashlight: energy moves from the battery to the bulb AND changes form to light. This is ___.

a) TRANSFER only b) TRANSFORMATION only c) BOTH

10. A buzzer: energy moves through the wires AND changes form to sound. This is ___.

a) TRANSFER only b) TRANSFORMATION only c) BOTH

PART 5 — Short Explanation

11. Choose one system (sound, light, heat, or electricity). Write a claim that energy transferred. Use the model: “I know ___ because ___.” Include the change you would observe as evidence.